

53 New Integer Schemes for 3×3 Matrix Multiplication with 23 Products

Reproducible report — logic repo, matmul track, 2026-07-03 (commits 45e5acc..93347f2). All artifacts referenced live in this repository; every claim has a mechanical check listed in §4.

1 Summary

We report **53 new schemes for multiplying two 3×3 matrices with 23 multiplications, with coefficients in $\{-1, 0, +1\}$** — valid over any commutative ring, the same object class as Laderman’s 1976 scheme. Each scheme is:

1. **verified mod 2** against all 729 Brent equations,
2. **pairwise inequivalent** under the full de Groote symmetry group ($\text{GL}(3, 2)^3 \rtimes S_3$, exact witness-or-refutation checks),
3. **inequivalent to every published scheme**: all 17,376 schemes of the Kauers–Heule–Seidl (HKS) database *and* the four classics (Laderman 1976, Smirnov 2013, Oh–Kim–Moon 2013, Courtois–Bard–Hulme 2011),
4. **lifted to integer coefficients** in $\{-1, 0, +1\}$ and verified **exactly over $\mathbb{Z}$** against all 729 integer Brent equations.

Discovery cost: the 53 came out of one **6-minute single-threaded run** of a neighborhood-walk pipeline (138 raw finds $\rightarrow$ 53 survived full-database novelty checking), plus ~ 40 minutes of certification compute. (All timings in this report are from one machine: a Mac mini with an Apple M4 Pro — 10 performance + 4 efficiency cores — and 64 GB RAM, macOS.) For scale: the HKS campaign that produced the 17,376-scheme database used ~ 35 CPU-years (methods differ; see §7 caveats).

The engine behind the discovery is a **native-ANF stochastic local search (SLS)**: the Brent system is kept as 729 cubic-XOR constraints over the 621 real variables instead of a $\sim 26,500$ -variable CNF, which extends the seeded-repair horizon by $\geq 5,000\times$ over yalsat-on-CNF and, with an exact GF(2) “tensor closure” move, solves 8/10 of the official HKS challenge-1 instances (yalsat’s published record: 5/10).

2 Background

A bilinear scheme for 3×3 matrix multiplication with r products is a triple of coefficient tensors α, β, γ :

$$M_m = \left(\sum_{a,b} \alpha_{ab}^{(m)} A_{ab} \right) \cdot \left(\sum_{c,d} \beta_{cd}^{(m)} B_{cd} \right), \quad m = 1, \dots, r$$

$$C_{pq} = \sum_m \gamma_{pq}^{(m)} M_m$$

Correctness is equivalent to the **Brent equations**; over GF(2):

$$\bigoplus_{m=1}^r \alpha_{ab}^{(m)} \wedge \beta_{cd}^{(m)} \wedge \gamma_{pq}^{(m)} = \delta_{bc} \delta_{ap} \delta_{dq} \quad \forall (a, b, c, d, p, q)$$

i.e. 729 cubic XOR equations over $27r$ variables ($r=23$: 621 vars; 27 equations have RHS 1 — the “type-3” / delta equations). Any integer scheme reduces mod 2 to a GF(2) scheme; conversely a GF(2) scheme *may* lift to signs ± 1 (HKS observe lifting rarely fails).

Known landscape: $r=23$ achievable (Laderman 1976); best lower bound 19 (Bläser 2003); $r=22$ **open in both directions**. Before HKS (SAT 2019 / J. Symbolic Computation 2021), only 4 inequivalent $\{-1, 0, 1\}$ schemes were known; their local-search campaign found >17,000 more, all published in the Linz database this report checks against.

3 The pipeline

Five stages; each has a tool in `matmul/` and an exact command in §4.

3.1 Native-ANF SLS engine (`src/anf.rs`, binary `anf`)

The Brent system is represented natively as cubic-XOR (“ANF”) constraints over the 621 real variables — no Tseitin auxiliaries. Flipping a variable touches exactly its 81 incident equations; a monomial toggles iff its two partner bits are 1, so incremental evaluation is $O(81)$ per flip (0.4–5M flips/s/core measured, ~ 10 M flips/s aggregate over 10 threads). Two policy regimes matter (both WalkSAT-family):

- close repair (many bits frozen at a known scheme): WalkSAT/SKC, noise 0.2, init density 0.25;
- pairing / from-scratch: probSAT, cb 2.5, **init density 0.10** ($\approx$ the free-support density of real completions).

The engine’s structural move is the **tensor closure**: the Brent system is tri-linear, and every equation contains exactly one variable-group of each tensor, so fixing two tensors decomposes the third into 9 independent $81 \times r$ GF(2) linear systems. A consistent single-tensor closure therefore *solves the instance outright*, and each closure call is monotone. It runs as an injected hook every N flips (`--closure-every`).

Baselines on this machine (the Mac mini defined in §1): kissat cannot solve the $r=23$ CNF (unknown at 60 s; it needs 41.6 s to prove even 2×2 -in-6 UNSAT), matching HKS’s finding that CDCL fails here. yalsat (the HKS solver, v1.0.1 built from source) solves seeded-repair CNFs at their published operating point (fix 414/621 bits: ~ 0.05 – 0.2 s) but **times out at 300 s at fix=300 and fix=250 — instances the native engine solves in 5 ms and 60 ms**. On the official challenge-1 instances (hardcoded pairing cores, no streamliners), native SLS + closure solves **8/10** (best two: 0.019 s and 0.069 s; the 0.069 s A-instance solution was additionally confirmed by planting our 621 bits as units into *their* CNF and running kissat: SATISFIABLE). yalsat’s published record on challenge 1 is 5/10 “in a few minutes”; same-machine yalsat probes: A and M time out at 300 s, 4-4-4-4-1 solves in 0.62 s (native: 0.019 s).

3.2 Discovery: neighborhood walk (`matmul/walk.py`)

HKS “method 2”, compounding: maintain a pool (24 verified seed schemes fetched from the Linz DB: 4 classics + 20 spanning 20 rank-pattern directories); repeatedly pick a pool scheme, freeze a random 300 of its 621 bits, let the engine complete the remaining 321 from a random start; canon-dedupe completions (sort the 23 summands); genuinely new schemes join the pool and become seeds themselves. Every accepted scheme is re-verified against the Brent equations by code independent of the search.

The committed run: `--minutes 6 --nfix 300 --runs 8 --rng 7`, single-threaded $\rightarrow$ **138 schemes, distinct after summand sorting, at ~ 3 s/scheme**, accelerating as the pool diversifies.

3.3 Exact equivalence (`matmul/equiv.py`)

De Groote’s symmetry group acts on schemes; “new” must mean “new modulo that group”. Key implementation fact: writing schemes as summands $(A, B, \tilde{C})$ with $\tilde{C} = \gamma^T$, the group action becomes the cyclic sandwich

$$(PAQ^{-1}, QBR^{-1}, R\tilde{C}P^{-1})$$

— and then every constraint “this summand maps to that one” is **linear** in the 27 unknown GF(2) bits of (P, Q, R) . Exact equivalence testing = backtracking over summand matchings (pruned by rank triples) + incremental GF(2) RREF + nullspace enumeration + invertibility + full multiset check; $\sim$ ms per pair. An

invariant fingerprint (multiset of per-summand sorted rank triples + pair-sum rank triples) prunes almost all pairs first.

Self-tests: 12 random group elements applied to Laderman are (a) still valid schemes and (b) recovered as equivalent-with-witness; Laderman vs Smirnov is refuted. On the 138 finds: **129 distinct classes** (9 pairs were product-reorderings/ G -images of each other).

3.4 Database novelty (matmul/novelty.py, matmul/dbcheck.py via agent)

Two independent layers:

1. **Rank-pattern absence.** The DB's 302 directory names encode per-summand rank types (legend constraint-solved from 20 known $\text{dir} \leftrightarrow \text{scheme}$ pairs; letters refine sorted rank triples by slot: $a=(1,1,1)$, $b/d/j=(1,1,2)$, $c/g/s=(1,1,3)$, $e/k/m=(1,2,2)$, $f=(1,2,3)$, $n=(2,2,2)$, $w=(2,2,3)$; unknown letters treated as wildcards — conservative). Rank patterns are G -invariants, so pattern-absence $\Rightarrow$ inequivalence. Control: Laderman's pattern is absent from all 302 found-scheme dirs — correct (HKS report their finds never reached Laderman's type).
2. **Full-database exact check.** The complete DB (single `schemes.tgz`, 43MB = **17,376 .tab files**, byte-validated against live fetches) was converted (γ -transposed), with **0/17,376 parse failures** — every file passes `verify_bits==0` — and 13/13 seed-anchor controls were recovered byte-identically. Per find: fingerprint comparison against all schemes in its compatible dirs, exact `equivalent()` on collisions. **Hardening:** all 17,376 DB schemes were fingerprinted and the surviving finds re-checked ($\times 6$ S_3 variants) against the *whole* database: **zero fingerprint matches anywhere**, making the verdicts independent of the legend.

Verdicts (`matmul/novelty_verdicts.csv`): **85 of 138 finds are equivalent to DB schemes** (expected — the walk is DB-seeded; witnesses recorded), **53 are new vs the entire database**, and (from §3.3's audit) inequivalent to the four classics and to each other: 53 finds = 53 classes.

3.5 Integer lifting (matmul/lift.py)

Hypothesis (HKS's): signs ± 1 exist on the same support. Encoded as **sign-SAT** rather than Gröbner bases: one sign bit per support coefficient; a covering term's sign is the XOR of its three sign bits; the integer Brent equation with k covering terms and RHS r becomes “exactly $(k - r)/2$ of the k term-bits are 1”; per-product scaling freedom is broken by fixing the first α - and β -support sign of each product. $\sim 2,000$ -clause CNFs; kissat solves each in milliseconds. Every lift is then verified **exactly over $\mathbb{Z}$** (integer arithmetic, all 729 equations) by code independent of the encoding.

Controls: all 4 classics lift and verify. Result: **53/53 lifted, zero failures**, outputs in `matmul/lifted/*.txt`.

4 Verification chain — check every claim yourself

All commands run from the repo root unless noted; `cd matmul` where shown. Expected wall time in brackets.

```
# 0. build + engine self-tests (incl. closure reconstruction,
#   incremental-vs-scratch equality, Laderman/Strassen embedding)  [~1 min]
cargo build --release --bin anf
cargo test --release --lib anf:           # expect: 7 passed

# 1. generator sanity: Strassen + Laderman verify, instance sizes  [~5 s]
cd matmul && python3 brent.py selftest

# 2. the 53 are valid + distinct-after-sorting                      [~10 s]
grep NEW novelty_verdicts.csv | cut -d, -f1 | sed 's|^|found/|;s|$.bits|' \
  | xargs cat | python3 canon.py 3 3 3 23 /dev/stdin
#   expect: 53 schemes read, 0 INVALID, 53 distinct after summand sorting

# 3. exact-equivalence machinery self-test + 53 = 53 classes      [~30 s]
```

```

python3 equiv.py selftest
grep NEW novelty_verdicts.csv | cut -d, -f1 | sed 's|^|found/|;s|$.bits|' \
| xargs python3 equiv.py classes
# expect: TOTAL de-Groote classes: 53

# 4. rank-pattern novelty vs the 302 DB dirs (weaker, no download) [~30 s]
python3 novelty.py db_rank_patterns.txt found/walk-00029.bits # etc.

# 5. FULL database check from primary sources [~15 min]
python3 dbcheck.py all
# fetch (43 MB schemes.tgz) -> convert all 17,376 .tab with a
# verify_bits==0 gate (expect 0 failures) -> controls (20 db seeds
# byte-identical + 4 classics equivalent) -> check every find against
# ALL DB fingerprints (x6 S3 variants; exact equivalent() on
# collisions; directory-scope-free, so independent of the rank-pattern
# legend). Ends with an automatic cross-check against the committed
# novelty_verdicts.csv: expect "MATCH" and 85 EQUIVALENT / 53 NEW.

# 6. integer lifting + exact Z-verification [~1 min]
python3 lift.py seeds/smironov.bits seeds/laderman.bits --outdir /tmp/ctl
grep NEW novelty_verdicts.csv | cut -d, -f1 | sed 's|^|found/|;s|$.bits|' \
| xargs python3 lift.py --outdir /tmp/lift53
# expect: "53 lifted, 0 not +-1-liftable"; each line Z-VERIFIED
# (the Z-check is an assertion: any failure aborts loudly)

# 7. challenge-1 spot-check (their exact CNF file, our scheme) [~1 min]
git clone https://github.com/marijnheule/matrix-challenges challenges
mkdir -p inst && python3 import_core.py \
challenges/challenge1/MM-23-2-2-2-2-A.cnf inst/core-A.freeze
../target/release/anf 3 3 3 23 --freeze-file inst/core-A.freeze \
--probsat --cb 2.5 --density 0.1 --seconds 60 --threads 10 --seed 3 \
--quiet | grep '^b ' > /tmp/solA.txt
python3 check_their_cnf.py challenges/challenge1/MM-23-2-2-2-2-A.cnf \
/tmp/solA.txt
# expect: "validated type-3 pairing (transpose_gamma=True)" on import,
# then: SATISFIED by our scheme
# (challenges/ and inst/ are gitignored -- external data + derived)

```

What is and isn't deterministic. Verification of the committed 53 (steps 0–6) is deterministic. *Discovery* (walk.py) and the challenge-1 solves are stochastic and timing-dependent: rerunning reproduces the *phenomena* (similar yields/times) but not bit-identical artifacts. kissat may return different sign models in step 6 across versions — any model it returns is then $\mathbb{Z}$ -verified, which is the claim that matters.

5 The 53 schemes

Files: mod-2 bit-vectors `matmul/found/walk-*.bits` (the 53 names are the NEW rows of `matmul/novelty_verdicts.csv`); signed integer forms `matmul/lifted/walk-*.txt`. Support = number of nonzero coefficients; for a $3 \times 3 \times 23$ scheme the naive addition count is exactly support – 55. Ours: support 149–164 (median 154) = **94–109 naive additions**; for reference Laderman = 153/98, Smirnov = 139/84, DB minimum = 139 (support percentiles of the full DB: $p_1 = 146$, median = 159 — our sparsest sits at the 3rd percentile; 22 of the 53 need fewer naive additions than Laderman, none beats the DB's sparsest). Rank-type multiset = per-summand sorted (rank α , rank β , rank γ) with multiplicities — the invariant that separates 51 of the 53 from the whole DB at the coarsest level.

Example (walk-00029, lifted; full file in `matmul/lifted/`):

```

M1 = (+a11 +a32) . (+b11 -b21 +b22)      C11 = +M1 +M10 -M12 +M13 +M17
M2 = (+a21 +a32) . (+b11 -b21 -b23)      C12 = +M1 -M4 -M12 +M13 +M20
... (23 products)                        ... (9 outputs)

```

Full table (53 rows)

scheme	support	rank-type multiset
walk-00028	157	111×12 112×2 113×3 122×2 222×4
walk-00029	156	111×11 112×3 113×3 122×2 222×4
walk-00034	157	111×12 112×2 113×3 122×2 222×4
walk-00035	150	111×14 112×3 113×1 122×1 222×4
walk-00036	151	111×15 112×2 113×1 122×1 222×4
walk-00039	163	111×10 112×7 122×2 222×4
walk-00040	160	111×10 112×7 122×2 222×4
walk-00046	149	111×13 112×6 222×4
walk-00047	153	111×16 112×3 222×4
walk-00048	158	111×16 112×3 222×4
walk-00055	157	111×12 112×5 122×2 222×4
walk-00056	157	111×14 112×3 113×1 122×1 222×4
walk-00058	160	111×10 112×7 122×2 222×4
walk-00060	155	111×8 112×8 113×1 122×2 222×4
walk-00063	155	111×14 112×5 222×4
walk-00064	155	111×15 112×4 222×4
walk-00066	151	111×9 112×8 113×1 122×1 222×4
walk-00072	151	111×12 112×3 113×3 122×1 222×4
walk-00075	154	111×8 112×6 113×4 122×1 222×4
walk-00076	152	111×13 112×6 222×4
walk-00077	150	111×15 112×4 222×4
walk-00081	150	111×14 112×3 113×1 122×1 222×4
walk-00082	152	111×13 112×6 222×4
walk-00083	152	111×13 112×6 222×4
walk-00084	149	111×13 112×6 222×4
walk-00085	153	111×16 112×3 222×4
walk-00091	156	111×10 112×4 113×2 122×3 222×4
walk-00092	153	111×9 112×7 113×1 122×2 222×4
walk-00093	153	111×8 112×9 113×1 122×1 222×4
walk-00094	151	111×9 112×5 113×4 122×1 222×4
walk-00100	160	111×11 112×6 122×2 222×4
walk-00105	151	111×11 112×8 222×4
walk-00106	161	111×9 112×5 113×3 122×2 222×4
walk-00107	149	111×10 112×5 113×3 122×1 222×4
walk-00108	158	111×11 112×3 113×3 122×2 222×4
walk-00115	154	111×14 112×3 113×1 122×1 222×4
walk-00116	155	111×14 112×2 113×1 122×2 222×4
walk-00120	152	111×14 112×3 113×1 122×1 222×4
walk-00121	164	111×8 112×5 113×4 122×2 222×4
walk-00122	163	111×9 112×5 113×3 122×2 222×4
walk-00125	151	111×13 112×4 113×1 122×1 222×4
walk-00127	155	111×15 112×4 222×4
walk-00136	154	111×13 112×6 222×4
walk-00138	154	111×11 112×4 113×3 122×1 222×4
walk-00141	160	111×12 112×2 113×3 122×2 222×4
walk-00143	152	111×13 112×4 113×1 122×1 222×4
walk-00144	150	111×13 112×2 113×3 122×1 222×4
walk-00145	149	111×12 112×3 113×3 122×1 222×4
walk-00149	156	111×14 112×3 113×1 122×1 222×4
walk-00150	159	111×14 112×3 113×1 122×1 222×4
walk-00151	150	111×14 112×3 113×1 122×1 222×4
walk-00157	151	111×11 112×8 222×4
walk-00159	150	111×13 112×2 113×3 122×1 222×4

(Note: the multiset shown is the coarse invariant; several schemes share it — they are separated by the finer pair-sum fingerprint and/or exact checks. All 53 are pairwise inequivalent.)

6 Secondary results

- **Challenge 1 (HKS): 8/10 official instances solved** (pairing cores, no streamliners; yalsat’s record 5/10). The two holdouts (M, 2-2-2-4-A) floor at best 3/729 after 600s×10 threads.
- **Path-space SLS (connection-method local search): built, verified correct, measured NEG-ATIVE** at equal budget vs assignment-space SLS (orders of magnitude, all regimes). Diagnosis: the connection objective collapses (a conflicted variable hides force1×force0 repairs) and the Brent matrix’s sharing is dense (81 equations/var) — path rerouting is myopic exactly where an assignment flip re-evaluates everything at once. `matmul/pathsls.py`, `doc/matmul_plan.md` R3c.
- **$r=22$ (challenge 4): open, probed.** Plain native attacks floor at 8/729. Drop-a-product repair floors at 1/729 — but the finisher (`matmul/finisher22.py`) proved every such floor-1 state violates exactly the type-3 equation whose sole cover was dropped, is rigid to radius 3 ($\sim 1.3M$ flip-sets), and fails per-tensor closure by exactly 1 row each: the floor-1 shell is a *seeding artifact*, not evidence about $r=22$. Campaign infrastructure: `matmul/campaign22.py` (checkpointed; wave 1 = 210 attacks logged).

7 Caveats and scope

- *Re-checked 2026-07-04*: the 53 were additionally tested against Perminov’s scheme collection (<https://github.com/dronperminov/FastMatrixMultiplication>, the source of the Dec-2025 additive-record schemes; 7 distinct rank-23 schemes, all themselves HKS-DB classes) — **zero matches; all 53 remain new**, with positive controls firing.
- “New” = inequivalent under de Groote symmetry to the 17,376 schemes in the Linz database snapshot (`schemes.tgz`, Last-Modified 2020-08-07) and the 4 classics. The database is the comprehensive public record of $\{-1, 0, 1\}^{3 \times 3 \times 23}$ schemes; we make no claim about unpublished or post-2020 private collections.
- Discovery-effort comparisons with HKS (35 CPU-years vs minutes) are indicative, not controlled: we *start from* their published schemes as seeds, they started from 4; their campaign also had to invent the methods. The like-for-like comparison is our seeded-repair and challenge-1 numbers vs yalsat on the same machine (§3.1).
- The de-Groote checker, DB converter, and lift verifier are our code; their self-tests and controls are listed in §4. The two strongest external anchors: our A-instance solution satisfies *HKS’s own CNF* under kissat, and the DB fingerprint hardening uses only rank arithmetic + our exact checker with published inputs.
- 51/53 schemes have four rank-(2,2,2) summands (like most HKS finds); none matches Laderman’s four-quadruple core type.

8 Pointers

- Plan/lab-notebook: `doc/matmul_plan.md` (every measurement, incl. negatives and retractions).
- Engine: `src/anf.rs`, `src/bin/anf.rs`.
- Tools (all in `matmul/`): `brent.py`, `sls.py`, `walk.py`, `canon.py`, `equiv.py`, `novelty.py`, `lift.py`, `import_core.py`, `check_their_cnf.py`, `dbcheck.py`, `campaign22.py`, `drop22.py`, `finisher22.py`, `pathsls.py`.
- Everything needed for §4 is in-repo except three things fetched from their public sources by the commands shown: the DB archive (`dbcheck.py fetch`), the matrix-challenges clone (step 7), and yalsat (<https://github.com/arminbiere/yalsat>) for the baseline timings.
- HKS: *Local Search for Fast Matrix Multiplication* (SAT 2019, arXiv:1903.11391); *New ways to multiply 3×3 -matrices* (J. Symbolic Computation 104, 2021).

- Scheme database: <https://algebra.uni-linz.ac.at/research/matrix-multiplication>.
- Challenges: <https://github.com/marijnheule/matrix-challenges>.